

Supplementary Material S2
MMAT Appraisal by Study and Domain

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements the figure in the manuscript presenting the aggregate distribution of MMAT judgements by domain (Section 2.6). Here the appraisal is reported study by study, separated into the 23 studies of the eligible corpus and the 7 studies of contextual evidence outside the eligible set.

In accordance with the manual of the Mixed Methods Appraisal Tool (MMAT) version 2018, which deliberately eliminated the numerical summative scoring present in earlier versions, no aggregate score per study is computed or reported. Each study was assigned to the corresponding design category and appraised with the five criteria proper to that category: criteria 3.1–3.5 for quantitative nonrandomised studies (Cat. 3), criteria 4.1–4.5 for quantitative descriptive studies (Cat. 4), and criteria 5.1–5.5 for mixed-methods studies (Cat. 5). The five judgement columns correspond to the five aggregated domains described in Section 2.6 of the manuscript: Rep. = Representativeness of the sample (3.1 / 4.2), Meas. = Adequacy of measurements (3.2 / 4.3), Part. = Partitioning protocol and sampling strategy (3.5 / 4.1), Conf. = Control of confounding factors and analysis (3.4 / 4.5), and Out. = Completeness of outcome data (3.3 / 4.4). Judgements are coded as Y = Yes, N = No, CT = Can’t Tell.

Section A. Eligible corpus (23 studies)

Of the 23 eligible studies, 22 are assignable to an MMAT design category (17 quantitative nonrandomised, 4 quantitative descriptive, 1 mixed methods) and 1 falls outside the instrument’s scope (systems design without supervised classification). The sum 17 + 4 + 1 + 1 recovers the 23 studies of the eligible corpus. Ahmad et al. (2024) was retracted on 17 August 2026 (Healthc Technol Lett 2026;13(1):e70095) and is therefore excluded from this appraisal.

#	Reference	Cat.	Rep.	Meas.	Part.	Conf.	Out.
<i>Category 3: Quantitative nonrandomised (17 studies)</i>							
1	Johnson et al. (2025)	3	Y	Y	Y	CT	Y
2	Shaik et al. (2025)	3	CT	Y	CT	N	Y
3	Radolaj & Martinović (2025)	3	Y	Y	Y	Y	Y
4	Washington et al. (2022)	3	Y	Y	Y	Y	Y
5	Sholikah et al. (2024)	3	CT	Y	CT	N	Y
6	Sariyanidi et al. (2025)	3	Y	Y	Y	Y	Y
7	Hassan et al. (2025)	3	Y	Y	Y	Y	Y
8	Alsubai (2025)	3	N	Y	N	N	Y
9	Rudovic et al. (2018)	3	Y	Y	Y	Y	Y
10	Cicceri et al. (2026)	3	Y	Y	Y	CT	Y
11	Rudovic et al. (2019)	3	Y	Y	Y	Y	Y
12	Aljbori et al. (2025)	3	N	Y	N	N	Y

Continued on next page

13	Vedantham et al. (2025)	3	N	Y	CT	N	Y
14	Hosney et al. (2024)	3	N	Y	N	N	Y
15	Leo et al. (2018)	3	Y	Y	Y	Y	Y
16	Hwang et al. (2024)	3	Y	Y	Y	Y	Y
17	Alam et al. (2025)	3	CT	Y	CT	N	Y
<i>Category 4: Quantitative descriptive (4 studies)</i>							
19	He et al. (2019)	4	Y	Y	Y	Y	Y
20	Reed et al. (2021)	4	Y	Y	Y	Y	Y
21	Dapogny et al. (2019)	4	Y	Y	Y	Y	Y
22	Akın-Bülbül et al. (2024)	4	Y	Y	Y	CT	Y
<i>Category 5: Mixed methods (1 study)</i>							
23	Lecciso et al. (2021)	5	Y	Y	Y	CT	Y
<i>Outside MMAT scope (1 study)</i>							
24	Almeida et al. (2019)	Systems design without supervised classification					

Section B. Contextual evidence outside the eligible set (7 studies)

The following seven studies do not form part of the eligible corpus and do not participate in any denominator of the manuscript. Their MMAT appraisal is documented here for completeness. Six are assignable to an MMAT design category (4 quantitative nonrandomised, 2 quantitative descriptive) and 1 falls outside the instrument’s scope. Each study’s reason for classification as contextual evidence is stated in Section 2.4 of the manuscript.

#	Reference	Cat.	Rep.	Meas.	Part.	Conf.	Out.	Reason
<i>Category 3: Quantitative nonrandomised (4 studies)</i>								
C1	El Rhatassi et al. (2024)	3	CT	Y	CT	N	Y	Adult participants
C2	El Rhatassi et al. (2025)	3	CT	Y	CT	N	Y	Adult participants
C3	Nagae & Lee (2022)	3	Y	Y	CT	CT	Y	EDA only (CE3)
C4	O’Sullivan et al. (2023)	3	Y	Y	Y	CT	Y	Speech only (CE3)
<i>Category 4: Quantitative descriptive (2 studies)</i>								
C5	Papaioannou et al. (2021)	4	Y	Y	Y	CT	Y	EEG only (CE3)
C6	Xu & Ju (2023)	4	Y	Y	Y	Y	Y	sMRI/VBM only (CE3)
<i>Outside MMAT scope (1 study)</i>								
C7	Levy et al. (2024)	Systems design; adult participants						

Notes on the appraisal.

1. The four retracted studies were excluded from the corpus and do not receive MMAT appraisal. Güngör et al. (2025), Mahmood et al. (2025), and Kumar et al. (2024) were excluded following the editorial status verification described in Section 2.7 of the manuscript. Ahmad et al. (2024) was retracted after corpus finalisation (17 August 2026, *Healthc Technol Lett* 2026;13(1):e70095) and is likewise excluded.
2. Studies classified as systems design (Almeida et al., 2019 in the eligible corpus; Levy et al., 2024 in the contextual set) do not correspond to any of the five MMAT design categories. Their quality is discussed narratively in Section 6.2 of the manuscript.
3. The judgement “CT” (Can’t Tell) in the Conf. domain for Johnson et al. (2025) and Lecciso et al. (2021) reflects that potential confounders were partially but not fully addressed; the same judgement for Cicceri et al. (2026) and Akın-Bülbül et al. (2024) reflects partial documentation of confounding control in clinical settings.
4. The negative judgements in Rep. and Conf. for Alsubai (2025), Aljbori et al. (2025), and Hosney et al. (2024) are consistent with their use of datasets from web repositories without clinical validation, as documented in the manuscript’s Table 18.
5. The MMAT does not capture risks specific to predictive modelling such as data leakage, image-level versus subject-level partitioning, or model calibration. This limitation is documented in Section 2.8 of the manuscript.